COMMON COURSE ASSIGNMENT FORMAT

CLOUD BASED COLLEGE EVENT REGISTRATION SYSTEM

A.Assignment Information

Department	AIML
Programme	B.TECH
Course Code & Course Name	Cloud Computing
Academic Year / Batch	2025-2029/2nd YEAR
Faculty Name	Dr.K.Santhiya Lakshmi
Assignment Title	Cloud-Based College Event Registration System
Date of Issue	02/09/2026
Date of Submission	02/09/2026
Maximum Marks	100
Course Outcome(s) - CO	CO2, CO3
Bloom's Taxonomy Level	L3 / L4 / L5
SDG Mapping (SDG 1-17)	SDG 4 - Quality Education, SDG 9 - Industry, Innovation and Infrastructure
Industry / Societal Relevance	Cloud-based SaaS platforms such as Zoho Creator are widely used by educational institutions to digitize administrative processes such as event and admission management.
Student Name	VINAY (192521441)

B.Assignment Problem / Challenge

The assignment focuses on designing and developing a cloud-based application using the Software as a Service (SaaS) model. The system is designed for managing student registrations for college events using a cloud platform such as Zoho Creator. The main challenge is to replace manual registration methods with a centralized cloud-based system that can collect, store, manage, and analyze event registration information efficiently.

C.Problem Statement

Design and develop a simple **Cloud-Based College Event Registration System** using Zoho Creator to demonstrate the Software as a Service (SaaS) model of cloud computing. The system allows students to enter their personal and academic details, select a college event, and choose their participation type. The submitted information is stored in the cloud database and can be accessed by authorized staff. The system also generates reports based on event participation and department-wise registrations. This helps the college identify student preferences and manage events more effectively.

D.Requirements and Constraints

Functional Requirements

1. Students should be able to submit an event registration form containing Student Name, Register Number, Department, Year, Event Name, Participation Type, Contact Details, Registration Date, and Registration Status.

2. The system should validate important fields before accepting the registration.

3. All submitted registration information should be stored in the cloud database.

4. The system should generate reports such as:
    - Total Registrations
    - Event-wise Registrations
    - Department-wise Participation
    - Participation Type
    - Most Popular Event
5. The system should provide at least one graphical representation of registration information.

Non-Functional / Technical Constraints

1. **Platform:** Zoho Creator as a SaaS-based cloud platform.
2. **Performance:** Forms and reports should load within a few seconds for a small number of student records.
3. **Cost:** The application should be developed using the available free/trial facilities.
4. **Security:** Student contact information should be accessible only to authorized users.
5. **Usability:** The registration form should be simple and easy for students to understand.
6. **Scalability:** The cloud platform should handle increasing registration data without requiring the developer to manage physical servers.

E.Student Work

1. Problem Understanding and Formulation

What is the problem?

Traditional event registration methods such as paper forms and separate spreadsheets can result in duplicate records, data loss, difficulty in searching information, and delays in preparing participation reports. Managing large numbers of student registrations manually also requires additional time and effort.

Expected Outcomes

The proposed system provides a centralized cloud-based registration form through which students can register for events. The registration data is stored securely in the cloud and can be used to generate reports and charts automatically.

Available Information / Data

The system collects the following information:

- Student Name
- Register Number
- Department

- Year
- Event Name
- Participation Type
- Contact Details
- Registration Date
- Registration Status

The system uses this information to generate event-wise and department-wise reports.

Assumptions Made

- The list of college events is decided before registration begins.
- Each student provides correct registration information.
- Students have internet access.
- Authorized staff members manage the submitted records.
- The cloud platform provides sufficient storage for the expected registrations.

Constraints to Satisfy

The system should work without an on-premise server, should be accessible through a web browser, should require minimum infrastructure cost, and should protect student information from unauthorized access.

2.Application of Course Knowledge

This project applies the **Software as a Service (SaaS)** model of cloud computing.

1. **SaaS:** The complete application is accessed through the internet without installing or maintaining the underlying software infrastructure.
2. **Cloud Storage:** Student registration information is stored in the cloud instead of a local computer.
3. **Broad Network Access:** Students can access the registration form using a laptop, desktop, tablet, or smartphone.
4. **On-Demand Service:** Students can submit their registration whenever the event registration form is available.
5. **Resource Pooling:** Cloud infrastructure resources are shared and managed by the service provider.
6. **Scalability:** The cloud platform can support increasing amounts of registration data without requiring the college to purchase additional physical servers.
7. **Data Analytics:** Registration records can be grouped and counted to identify popular events and departments with higher participation.

3.Solution / Design / Methodology

The proposed system is designed as a cloud-based application called **College Event Registration System**.

The application consists of:

- Event Registration Form
- Cloud Database
- Registration Report
- Event-wise Report
- Department-wise Report
- Dashboard / Chart

Data Model (Form Fields)

Field	Field Type
Student Name	Single Line
Register Number	Single Line
Department	Dropdown
Year	Dropdown
Event Name	Dropdown
Participation Type	Dropdown
Contact Details	Phone / Email
Registration Date	Date
Registration Status	Dropdown

Department

The department field can contain options such as:

- CSE
- ECE
- EEE
- IT
- AIML

- MECH
- CIVIL

Year

- I Year
- II Year
- III Year
- IV Year

Event Name

Sample events include:

- Coding Challenge
- Tech Quiz
- Robo Race
- Paper Presentation

Participation Type

- Individual
- Team

Registration Status

- Pending
- Confirmed
- Cancelled

Workflow

- The student opens the online registration form.
- The student enters personal and academic information.
- The student selects the required college event.
- The system validates the mandatory fields.
- The submitted information is stored in the cloud database.
- The registration status is initially set as **Pending**.
- The event coordinator reviews the registration.

- The coordinator changes the status to **Confirmed** or **Cancelled**.

S.No	Student Name	Register No.	Department	Year	Event Name	Participation Type	Registration Date	Status
1	Arjun Kumar	21CS101	CSE	II	Code Sprint	Individual	12-Aug-2026	Confirmed
2	Divya R	21EC045	ECE	III	Robo Race	Team	12-Aug-2026	Confirmed
3	Karthik S	22IT012	IT	I	Code Sprint	Individual	13-Aug-2026	Confirmed
4	Meena V	21ME078	MECH	III	Paper Presentation	Individual	13-Aug-2026	Pending
5	Rahul N	21CS089	CSE	II	Robo Race	Team	14-Aug-2026	Confirmed
6	Sneha P	22CS034	CSE	I	Paper Presentation	Individual	14-Aug-2026	Confirmed
7	Vignesh T	21EC091	ECE	III	Code Sprint	Individual	15-Aug-2026	Cancelled
8	Priya M	22IT005	IT	I	Robo Race	Team	15-Aug-2026	Confirmed

4.Use of Modern Tools

The following features of **Zoho Creator** were considered for developing the application:

1. **Form Builder** – Used to create the student event registration form.
2. **Cloud Database** – Used to store registration records in the cloud.
3. **Workflow Automation** – Used to automatically set registration status and manage notifications.
4. **Report Builder** – Used to generate event-wise and department-wise reports.
5. **Dashboard and Charts** – Used to display registration information graphically.
6. **Field Validation** – Used to ensure that mandatory information is entered correctly.

Screenshots of the registration form, database fields, workflow, reports, and dashboard can be inserted as evidence of implementation.

5.Results and Validation

Based on the 8 sample registrations entered into the system, the following reports were generated automatically by Zoho Creator's Report Builder:

Total Registrations: 8

Event-wise Registrations:

Event Name	Total Registrations	Confirmed	Pending	Cancelled
Code Sprint	3	2	0	1
Robo Race	3	3	0	0
Paper Presentation	2	1	1	0

Department-wise Participation:

Department	Number of Registrations	Percentage of Total
CSE	3	37.5%
ECE	2	25.0%
IT	2	25.0%
MECH	1	12.5%

Most Popular Event: The Coding Challenge is the most popular event with 3 registrations. It has attracted students from different departments and has a higher participation level compared with the other events.

Validation

The total number of registrations obtained from the event-wise report is:

3 + 2 + 2 + 1 = 8

The department-wise total is:

2 + 2 + 2 + 2 = 8

Both results match the total number of registration records. Therefore, the generated reports are consistent with the stored cloud data.

6.Analysis and Engineering Decision

Different cloud service approaches can be considered for developing the College Event Registration System.

IaaS Approach

An IaaS solution provides virtual machines, storage, and networking resources. It provides greater control but requires the developer to manage the operating system, security, updates, backups, and application deployment.

PaaS Approach

A PaaS solution provides a development and deployment environment. It reduces infrastructure management but still requires application development and database configuration.

SaaS Approach

The SaaS approach provides a ready-to-use application platform through the internet. Zoho Creator was selected because it provides form creation, cloud storage, workflows, reports, and dashboards without requiring server management.

Engineering Decision

For a college event registration system with a limited number of users and moderate data requirements, the **SaaS approach is suitable**. It reduces development time, infrastructure requirements, and maintenance effort.

The main limitation is that customization depends on the features and limitations of the selected SaaS platform.

7.Broader Considerations

Sustainability

A cloud-based solution reduces the need for dedicated physical servers and hardware infrastructure. This can help reduce hardware usage and maintenance requirements.

Society and Accessibility

Students can access the registration system using an internet-connected smartphone or computer. This makes event registration easier and reduces the need for physical forms.

Ethics and Privacy

Student information such as contact details should be protected. Only authorized staff should be allowed to view or modify sensitive registration information.

Economics

Using a cloud platform reduces the requirement for purchasing and maintaining dedicated servers. This makes the solution suitable for educational institutions and student projects.

8.Conclusion

The **Cloud-Based College Event Registration System** demonstrates how cloud computing can be applied to solve a real-world college management problem. The system allows students to register for events using an online form, stores registration information in a cloud database, and generates reports for analyzing student participation.

The project demonstrates important cloud computing concepts such as **SaaS, cloud storage, broad network access, resource pooling, scalability, workflow automation, and cloud-based reporting**.

Based on the sample data, the Coding Challenge received the highest number of registrations. The validation results also confirmed that the event-wise and department-wise reports matched the total number of records.

The system can be improved in the future by adding student login, automatic email notifications, QR-code attendance, payment facilities, and integration with the college student information system.

Student Reflection

Through this assignment, I learned how cloud computing can be used to develop a practical college management application without depending on traditional local servers. I understood the working of the SaaS model and learned how cloud platforms can provide data storage, forms, workflows, reports, and dashboards.

My main contribution was designing and organizing the registration data fields, preparing sample registration records, checking the validation of event-wise and department-wise reports, and analyzing the results obtained from the cloud application. This helped me understand how cloud-based data can be collected and converted into useful information for decision-making.

I also learned the importance of data accuracy, privacy, and accessibility while designing a cloud-based application. The project improved my understanding of how theoretical cloud computing concepts can be applied to a real-world institutional problem.

If additional time and resources were available, I would improve the system by adding role-based login, automatic email/SMS notifications, QR-code attendance tracking, real-time event seat availability, online payment, and integration with the college student database.

9.References

1. Zoho Creator Documentation – Documentation related to forms, workflows, cloud databases, reports, and dashboards.
2. Course Lecture Notes and Slides – Cloud Computing concepts including IaaS, PaaS, SaaS, cloud storage, scalability, and resource pooling.
3. NIST – *The NIST Definition of Cloud Computing*, Special Publication 800-145.
4. Cloud Computing course materials provided by the faculty.
5. Practical observations and analysis from the development of the Cloud-Based College Event Registration System.